# Singapore School Kinderland

## Continual Assessment 1

## 2026 - 2027

## Secondary 2

## Computer Science

Part 1 (Answer Key)

Date: _____ Name: **TEACHER ANSWER KEY**

**Instructions:**

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

| | |
|---|---|
| **Total marks** | **35** |
| **Part 1** | **30** |
| **CW** | **5** |
| Marks Obtained | **30 / 30** |

## Section A: Multiple Choice Questions (MCQ) (6x1=6 marks)

**For each question 1-6, four options are given. Circle the correct option i, ii, iii or iv.**

**1.** Which digits are used in the Binary (Base-2) number system?

**i. 0 and 1 only** **ii.** 0 to 9
**iii.** 1 and 2 only **iv.** 0 to 9 and A to F

**2.** What is the Denary (Base-10) value of the 4-bit binary number 1010?

**i.** 8 **ii. 10**
**iii.** 12 **iv.** 15

**3.** In the Hexadecimal (Base-16) system, which Denary number is represented by the letter F?

**i.** 10 **ii.** 12
**iii.** 14 **iv. 15**

**4.** How many binary bits are grouped together to represent ONE Hexadecimal digit?

**i.** 2 bits **ii. 4 bits (1 nibble)**
**iii.** 8 bits **iv.** 16 bits

**5.** In Binary Addition, what is the result of adding 1 + 1?

**i.** 2 (Carry 0) **ii.** 0 (Carry 0)
**iii. 0 with Carry 1 (10 in binary)** **iv.** 1 with Carry 1 (11 in binary)

**6.** What is the largest Denary number that can be stored in an 8-bit byte (1111 1111)?

**i.** 128 **ii. 255**
**iii.** 256 **iv.** 512

## Section B: Binary and Denary Conversions (4x2=8 marks)

**7. Use the 8-bit place values (128, 64, 32, 16, 8, 4, 2, 1) to complete the conversions below:**

| No. | Conversion Task & Working Space | Final Answer |
|---|---|---|
| i. | Convert 8-bit Binary **0010 1101** to Denary (Base-10):<br>**Working: 32 + 8 + 4 + 1 = 45** | **45** |
| ii. | Convert 8-bit Binary **1001 0110** to Denary (Base-10):<br>**Working: 128 + 16 + 4 + 2 = 150** | **150** |
| iii. | Convert Denary **75** into an 8-bit Binary number:<br>**Working: 64 + 8 + 2 + 1 = 75** | **0100 1011** |
| iv. | Convert Denary **196** into an 8-bit Binary number:<br>**Working: 128 + 64 + 4 = 196** | **1100 0100** |

## Section C: Binary and Hexadecimal Conversions (4x2=8 marks)

**8. Convert between Binary and Hexadecimal using 4-bit groups / nibbles (8, 4, 2, 1):**

| No. | Conversion Task & Working Space | Final Answer |
|---|---|---|
| i. | Convert 8-bit Binary **1010 0101** to Hexadecimal:<br>**Working: 1010 = 10 (A), 0101 = 5** | **A5** |
| ii. | Convert 8-bit Binary **1111 0011** to Hexadecimal:<br>**Working: 1111 = 15 (F), 0011 = 3** | **F3** |
| iii. | Convert Hexadecimal **4C** into an 8-bit Binary number:<br>**Working: 4 = 0100, C (12) = 1100** | **0100 1100** |
| iv. | Convert Hexadecimal **9E** into an 8-bit Binary number:<br>**Working: 9 = 1001, E (14) = 1110** | **1001 1110** |

## Section D: Binary Addition (8-Bit Numbers) (4x2=8 marks)

**9. Add the following 8-bit binary numbers. Show your Carry row and Final 8-bit Sum clearly:**

**Binary Addition Rules:** 0 + 0 = 0 | 0 + 1 = 1 | 1 + 0 = 1 | **1 + 1 = 0 (Carry 1)** | **1 + 1 + 1 = 1 (Carry 1)**

| No. | 8-Bit Binary Addition Problem | Carry & Final Sum |
|---|---|---|
| i. | `  0 0 1 0 0 1 0 1`<br>`+ 0 0 0 1 0 0 1 0` | **Sum: 0011 0111**<br>**(Carry: 0000 0000)** |
| ii. | `  0 1 0 0 1 0 1 1`<br>`+ 0 0 1 0 0 1 1 0` | **Sum: 0111 0001**<br>**(Carry: 0001 1100)** |
| iii. | `  0 1 0 1 1 1 0 0`<br>`+ 0 0 1 1 1 0 1 0` | **Sum: 1001 0110**<br>**(Carry: 0111 0000)** |
| iv. | `  0 1 1 0 1 1 0 1`<br>`+ 0 1 0 1 0 1 1 1` | **Sum: 1100 0100**<br>**(Carry: 0111 1110)** |